MILITARY SPECIFICATION

TRAINER, TURRET: IMPROVED TOW VEHICLE

This specification is approved for use by US Army Tank-Automotive Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers turret trainer for M901 improved TOW vehicle (ITV). Trainer includes turret, armament, fire control, support items, and mounting frame.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

TT-E-529	- Enamel, Alkyd, Semi-gloss.
TT-P-664	- Primer, Coating, Synthetic, Rust-inhibiting, Lacquer-resisting.

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MIL-W-5044	- Walkway Compound, Nonslip, and Walkway Matting, Nonslip.
MIL-W-5050	- Walkway, Coating and Matting, Nonslip, Aircraft, Application of (Asg).
MIL-C-8507	- Coating, Wash Primer (Pretreatment) for Metals, Application of (for Aeronautical Use) (Asg).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Tank-Automotive Command, ATTN: DRSTA-GSS, Warren, MI 48090, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-T-23991	- Training Devices, Military, General Specification for.
MIL-E-52798	- Enamel, Alkyd, Camouflage.
MIL-E-52835	- Enamel, Modified Alkyd, Camouflage, Lusterless.
MIL-V-62347(AT)	- Vehicle, Combat, Anti-tank, Improved TOW Vehicle, (Less TOW Weapon), M90l.

STANDARDS

FEDERAL

FED-STD-H28/2	- Screw Thread Standards for Federal Services.

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-454	- Standard General Requirements for Electronic Equipment.
MIL-STD-461	- Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference.
MIL-STD-462	- Electromagnetic Interference Characteristics, Measurement of.
MIL-STD-889	- Dissimilar Metals.
MIL-STD-1472	- Human Engineering Design Criteria for Military Systems, Equipment and Facilities.

2.1.2 Other Government documents, drawings, and publications. The following other Government documents and drawings form a part of this specification to the extent specified herein.

DRAWINGS

ARMY

10189465	- Battery, Missile Guidance Set.
10861607	- Hook.
10890323	- Pin.
12265500	- System Schematic, ITV, M90l.
12265568	- Marker.
12265600	- Carrier, Personnel, Armored, Full Tracked Assembly.
12265611	- Marker.
12265621	- Kit, Modification, Carrier, Personnel, Armored, Full Tracked.
12265635	- Platform.
12265714	- Kit, Modification, Carrier, Personnel, Armored, Full Tracked.
12265723	- Kit, Electrical, M901 Vehicle.
12265800	- Cupola Assembly, Turret (Turret Kit).

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12266001	- Erector-launcher Arm Assembly (Armament Kit) (Installation).
12266086	- Electrical Installation, ITV.
12266098	- Erection Arm.
12266278	- Launcher.
12266400	- Sight/Optics Fire Control Installation (Fire Control Kit).
12266810	- Guard.
12277420	- Support Assembly.
12277600	- M113 Modification Kit.
12277611	- Simulator Assembly - Boresight and Remote Arming Device.
12277801	- Trainer Assembly.
12277803	- Main Frame Weldment Trainer (ITV).
12277804	- Ladder Assembly.
12277809	- Cargo Door.
12277862	- Installation, Electrical TOW Systems Trainer.
12278785	- Decal.

(Copies of specifications, standards, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the acquisitioning activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 First article (initial production). Trainers to be furnished under this specification shall be represented by samples taken from initial production after award of contract. Samples shall be produced using tools and equipment that shall be used for manufacturing in quantity. Samples shall withstand initial production inspection specified herein (see 6.3).

3.2 Materials. Materials shall be as specified herein and in referenced specifications, standards and drawings. Material shall be free from defects which adversely affect performance or serviceability of the finished product (see 6.5).

3.3 Design and construction. Unless otherwise specified herein, trainer design and construction shall conform to Drawing 12277801, to drawings listed in table I, and to such other drawings, specifications, and standards as may be applicable. Parts with same part number shall be interchangeable as to fit, form, and function. Electronic parts and assemblies shall be interchangeable in accordance with requirement 7, MIL-STD-454.

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TABLE I. Turret trainer and subassemblies.

Drawing no.	Subassembly
10189465	Battery, Missile Guidance Set.
10861607	Hook.
10890323	Pin.
12265500	System Schematic, ITV, M901.
12265568	Marker.
12265611	Marker.
12265621	Kit, Modification, Carrier, Personnel, Armored Full Tracked.
12265635	Platform.
12265714	Kit, Modification, Carrier, Personnel, Armored Full Tracked.
12265723	Electrical Kit, M901 Vehicle.
12265800	Cupola Assembly, Turret (Turret Kit).
12266001	Erector-launcher Arm Assembly (Armament Kit) (Installation).
12266086	Electrical Installation, ITV.
12266098	Erection Arm.
12266278	Launcher.
12266400	Sight/Optics Fire Control Installation (Fire Control Kit).
12266810	Guard.
12277420	Support Assembly.
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12277611	Simulator Assembly - Boresight and Remote Arming Device.
12277801	Trainer Assembly.
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12277804	Ladder Assembly.
12277809	Cargo Door.
12277862	Installation, Electrical TOW Systems Trainer.
12298785	Decal.

3.3.1 Government furnished equipment. Government furnished equipment (GFE) will be as specified in applicable contract or purchase order (see 6.2).

3.3.2 Trainer. Unless otherwise specified herein, trainer and subassemblies shall meet the requirements of MIL-T-23991.

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3.3.3 Main frame weldment. Main frame weldment shall conform to Drawing 12277803 and shall provide mounting for turret cupola assembly Drawing 12265800; erector-launcher arm assembly Drawing 12266001; sight and optics fire control installation Drawing 12266400; armored full track personnel carrier modification kit Drawing 12265621; and trainer support items specified herein.

3.3.3.1 Walls. Main frame weldment shall be equipped with removable wooden walls at front, and on right and left sides.

3.3.3.2 Level sensors. Main frame weldment shall be equipped with level sensors (inclinometers) to indicate longitudinal and lateral cant angles between 0 and 10 degrees (°) in each direction on each axis.

3.3.3.3 Electrical junction box. Main frame weldment shall be equipped with electrical junction box including cables, connectors, switches, and equipment arranged in circuits conforming to Drawing 12277862, and requirement 1, MIL-STD-454. Weldment shall be direct current (dc) ground. Operating power shall be converted from 208/220/440 volts root-mean-square (Vrms), three phase, 60 Hertz (Hz), 30 KVA maximum, alternating current (ac) power source.

3.3.3.4 Power cable. Main frame weldment shall be equipped with electrical power cable not less than 20 feet in length, designed in accordance with MIL-STD-454.

3.3.4 Turret cupola assembly. Trainer shall be equipped with turret cupola assembly conforming to Drawing 12265800. Turret shall mount to main frame weldment in manner specified on Drawing 12265600.

3.3.5 Guided missile launcher. Turret shall be equipped with erection arm conforming to Drawing 12266098 and launcher conforming to drawing 12266278. Installation shall conform to Drawing 12266001.

3.3.6 Guided missile mount kit. Trainer shall be equipped with the following removable items conforming to Drawing 12265600:

Missile storage racks
Launch tube mounting assembly
Traversing unit mounting assembly
Tripod mounting assembly

3.3.7 Squad leader's periscope. Trainer shall be equipped with squad leader's periscope specified on Drawing 12265600.

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3.3.8 M26 periscope. Trainer shall be equipped with M26 periscope specified on Drawing 12265600.

3.3.9 M60 machine gun. Trainer shall be equipped with M60 machine gun (see table I) mounted on cupola with pintle pin mount and traversing rail. Machine gun mount shall lock at any point along traversing rail. When in stow position, machine gun shall not interfere with operation of other subassemblies of trainer.

3.3.10 Smoke grenade launcher arm/fire box. Trainer shall be equipped with GFE smoke grenade launcher arm/fire box.

3.3.11 Cargo door. Trainer shall be equipped with simulated cargo door, including door support, closing latch, two position bolt plate, and door bolt conforming to Drawing 12265600. Door may be constructed of lightweight materials suitable for training purposes. Door shall be equipped with means to hold it open in either one of two positions, LOAD or FULL.

3.3.12 Communications. Trainer shall be equipped with communications equipment as specified on Drawing 12277862. Five stations shall be provided for on-board communications.

3.3.13 Welding. Welding procedures shall conform to applicable drawing, specification, or standard. Misalignment and dimensional variations shall be within limits specified on applicable drawing (see 6.2).

3.3.14 Standard parts. Military standard parts shall be incorporated whenever possible. Commercial standard parts may be used provided they are interchangeable with military standard parts without modification.

3.3.15 Threaded parts. Screw threads of the form, number per inch, and class specified on the applicable drawing or military standard shall be in accordance with FED-STD-H28/2.

3.3.15.1 Locking devices. Lock washers, self-locking nuts, safety wires, or other approved locking devices shall be incorporated where specified or where required to prevent loosening of components.

3.3.16 Dissimilar metals. Dissimilar metals shall be protected from galvanic corrosion in accordance with requirements of MIL-STD-889.

3.4 Performance.

3.4.1 Strength and rigidity. Fully assembled trainer with a training class onboard weighing 1100 pounds (lbs) shall conform to requirements specified herein while canted to any longitudinal or lateral angle not more than 10° from normal. Canting trainer to 10° angle shall not cause permanent distortion of main frame weldment nor impair operation of subassemblies.

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3.4.2 Lift eye hoist. Lift eyes, when used together for hoisting, shall support weight of assembled trainer. Trainer shall withstand such hoisting with no impairment of operation nor permanent distortion.

3.4.3 Fork lift transport. Lift fork bearings, when used together for lifting, shall support weight of assembled trainer. Trainer shall withstand such lifting with no impairment of operation nor permanent distortion.

3.4.4 Towing attachments. Towing attachments shall withstand pull required to move assembled trainer on casters over smooth, hard surface. Trainer shall withstand such towing with no impairment of operation nor permanent distortion.

3.4.5 Casters. Casters shall support assembled trainer and shall roll and turn as required when trainer is pushed or towed over smooth, hard surface. Trainer shall withstand moving or standing on casters with no impairment of operation nor permanent distortion.

3.4.6 Stow supports and brackects. Stow supports and brackets shall receive launcher and support its weight in stowed position.

3.4.7 Seats. Seats shall rigidly support crew members.

3.4.8 Screw jacks and jack stands. Screw jacks and jack stands shall support assembled trainer, with crew, at any longitudinal or lateral angle not more than 10° from normal. Trainer shall withstand such displacement with no impairment of operation nor permanent distortion.

3.4.9 Access ladder. Access ladder shall rigidly support personnel during ascent and descent and shall be easily removed and securely stowed.

3.4.10 Stanchions and safety chains. Removable stanchions and safety chains shall be installed around the perimeter of the trainer upper-deck with chains 42 inches and 21 inches above deck surface. Each section of chain shall support a leaning weight of 200 pounds.

3.4.11 Electrical operation. Trainer subassemblies shall operate with 28 volt (V) direct current (dc) (+4 -6 Vdc), 200 ampere (A) circuit.

3.4.12 Current surge. 28 Vdc, 200 A circuits shall withstand current surge of 700 A for one second with no impairment of operation.

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3.4.13 Steady state voltage. Steady state output ripple voltage shall be not more than 4 V peak-to-peak over frequency range of 50 Hertz (Hz) to 200 kHz.

3.4.14 Electromagnetic interference characteristics. Electromagnetic interference characteristics of trainer shall conform to classes A-3 and C-1 of MIL-STD-461.

3.4.15 Safety interrupts and override controls. Safety interrupts and override controls shall govern operation of turret and missile launcher as specified in table II.

3.4.15.1 Launcher fire no-fire zones. Safety interrupts shall govern missile launching as specified in tables II and III.

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TABLE II. Safety interrupts and override conditions.

CONDITION / ACTION	LAUNCHER - ERECT					LAUNCHER - LOAD					LAUNCHER - STOWED				
	DRIVER HATCH	GUNNER HATCH	CARGO HATCH (LOAD)	CARGO HATCH	AZ/EL FIRE ZONE	DRIVER HATCH	GUNNER HATCH	CARGO HATCH (LOAD)	CARGO HATCH	AZ/EL FIRE ZONE	DRIVER HATCH	GUNNER HATCH	CARGO HATCH (LOAD)	CARGO HATCH	AZ/EL FIRE ZONE
	OP CL	OP CL	OP	OP CL	NO	OP CL	OP CL	OP	OP CL	NO	OP CL	OP CL	OP	OP CL	NO
AUDIBLE ALARM	● o	● o	●	● o	●	● o	● o	●	● o	o	● o	● o	●	● o	o
ERECTION DRIVE															
STOW TO ERECT	o o	o o	⊘	⊘ o	o	o o	o o	⊘	o o	o	● ●	⊘ ●	o	o ●	o
ERECT TO STOW	● ●	● ●	⊘	⊘ ●	●	o o	o o	o	o o	o	o o	o o	o	o o	o
ERECT TO LOAD	● ●	⊘ ●	●	● o	●	o o	o o	o	o o	o	o o	o o	o	o o	o
LOAD TO ERECT	o o	o o	o	⊘ o	o	● ●	● ●	●	● ●	●	o o	o o	o	o o	o
MISSILE LOAD	o ●	⊘ ●	●	● ⊘	o	● ●	● ●	●	● o	●	o o	o o	o	o o	o
MISSILE SELECT	● ●	● ●	●	● ●	⊘	⊘ ⊘	⊘ ⊘	⊘	⊘ ⊘	⊘	o o	⊘ ⊘	⊘	⊘ ⊘	⊘
MISSILE ARM	● ●	● ●	●	● ●	⊘	⊘ ⊘	⊘ ⊘	⊘	⊘ ⊘	⊘	o o	⊘ ⊘	⊘	⊘ ⊘	⊘
MISSILE FIRE	⊘ ●	⊘ ●	⊘	⊘ ●	⊘	⊘ ⊘	⊘ ⊘	⊘	⊘ ⊘	⊘	⊘ ⊘	⊘ ⊘	⊘	⊘ ⊘	⊘
MISSILE STOW	● ●	● ●	⊘	⊘ ●	o	● ●	● ●	●	● o	o	o o	o o	o	o o	o
CONTROL PANEL LAMPS															
DRIVER'S HATCH	● o	o o	o	o o	o	● o	o o	o	o o	o	● o	o o	o	o o	o
GUNNER'S HATCH	o o	● o	o	o o	o	o o	● o	o	o o	o	o o	● o	o	o o	o
AZ-LOAD	o o	o o	o	o o	o	● ●	● ●	●	● ●	●	o o	o o	o	o o	o
EL-LOAD	o o	o o	o	o o	o	● ●	● ●	●	● ●	●	o o	o o	o	o o	o
READY TO LOAD	o o	o o	o	o o	o	● ●	● ●	●	● ●	●	o o	o o	o	o o	o
STOW	o o	o o	o	o o	o	o o	o o	o	o o	o	● ●	● ●	●	● ●	●
CARGO HATCH	o o	o o	●	● o	o	o o	o o	●	● o	o	o o	o o	●	● o	o
FIRE INTERRUPT	● o	● o	●	● o	●	● ●	● ●	●	● ●	●	● ●	● ●	●	● ●	●
AZIMUTH DRIVE	⊘ ●	⊘ ●	o	o ●	●	⊘ ⊘	⊘ ⊘	⊘	⊘ ⊘	⊘	o o	⊘ o	o	o o	o
ELEVATION DRIVE	⊘ ●	⊘ ●	●	● ●	●	⊘ ⊘	⊘ ⊘	⊘	⊘ ⊘	⊘	o o	⊘ o	o	o o	o

Switch open, lamp extinguished, alarm silent = o
Switch closed, lamp lit, alarm sounds = ●
May be overridden = ⊘
No override = ⊘
OP = open
CL = closed
NO = no-fire

NOTE: When two or more conditions exist simultaneously, open switches shall govern circuits.

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TABLE II. Safety interrupts and override conditions - Continued.

CONDITION / ACTION	LAUNCHER - ERECT AZ-EL No-fire Zones		
	AZ 12.5°-58° EL-22°	AZ 118°-233° EL-13°	AZ 310°-347.5° EL-22°
AUDIBLE ALARM	●	●	●
ERECTION DRIVE			
STOW TO ERECT	o	o	o
ERECT TO STOW	o	o	o
ERECT TO LOAD	o	o	o
LOAD TO ERECT	o	o	o
MISSILE LOAD	o	o	o
MISSILE SELECT	%	%	%
MISSILE ARM	%	%	%
MISSILE FIRE	%	%	%
MISSILE STOW	o	o	o
CONTROL PANEL LAMPS			
DRIVER'S HATCH	o	o	o
GUNNER'S HATCH	o	o	o
AZ-LOAD	o	o	o
EL-LOAD	o	o	o
READY TO LOAD	o	o	o
STOW	o	o	o
CARGO HATCH	o	o	o
FIRE INTERRUPT	●	●	●
AZIMUTH DRIVE	●	●	●
ELEVATION DRIVE	●	●	●

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TABLE III. Launcher fire/no-fire zones.

Turret azimuth Relative to trainer (degrees ± 3.5°)	Launcher elevation angles	
	Fire zones	No-fire zones
347.5 - 12.5	34° to -30°	
12.5 - 58	34° to -22°	-22° to -30°
58 - 118	34° to -30°	
118 - 233	34° to -13°	-22° to -30°
233 - 310	34° to -30°	
310 - 347.5	34° to -22°	-22° to -30°

3.4.15.2 Firing position indication. When erection arm and launcher assembly are in firing position and all hatches are closed, all indicator lamps shall be extinguished.

3.4.15.3 Loading position indication. When erection arm and launcher assembly are in loading position, AZ LOAD, EL LOAD, and READY TO LOAD indicator lamps shall light.

3.4.15.4 Stow position indication. When erection arm and launcher assembly are in stow position, STOW indicator lamp shall light.

3.4.16 Intercom stations. Each intercom station shall permit intercommunication with all other intercom stations through 360° rotation of turret

3.4.16.1 Hydraulic pump noise. System shall be monitored for noise that interferes with intercommunications when weapons station hydraulic pump is cycled. Noise level shall be measured across a 10.5 kilohm resistor, connected across the audio output terminals, and shall be less than 350 millivolts AC rms as the pump cycles.

3.4.17 M60 machine gun mount. Effort required to move gun along traversing rail shall measure not more than 15 pounds after breakaway.

3.4.17.1 Locking device. Effort required to move gun, when locked in place on traversing rail, shall measure not less than 45 pounds.

3.4.18 Azimut drive speeds - turret. Azimuth drive shall provide speed control over continuous rotation in two ranges, track and slew.

3.4.18.1 Slew rate. Azimuth drive shall rotate turret at 35° per second in either direction.

3.4.18.2 Tracking rate. Azimuth drive shall rotate turret at 0.1 milliradians (mrad) per second in either direction.

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3.4.19 Elevation drive speed - launcher. Elevation drive shall move launcher at not less than 12° per second between minus 25° and 30° elevation. Total travel shall range between 30° ± 1° depression and 34° elevation.

3.4.19.1 Elevation minimum speed. Elevation drive speed shall be adjustable to a minimum rate of not more than 0.1 mrad per second.

3.4.20 Erection drive speed - launcher. Erection drive shall move launcher from stowed position to erect-locked position in not more than 8 seconds.

3.5 Finish.

3.5.1 Main frame weldment. Main frame weldment shall be primed in accordance with TT-P-664, composition G, and shall be painted in accordance with TT-E-529, class A or B. Color shall be forest green as specified in MIL-E-52798 and MIL-E-52835.

3.5.2 Walkways. Exterior and interior walkways shall be primed in accordance with MIL-C-8507 and treated with non-skid material conforming to MIL-W-5044, type II, class 2. Non-skid material shall be applied in accordance with MIL-W-5050, color shall be as specified in 3.5.1.

3.6 Marking.

3.6.1 Identification. Trainer shall be marked in accordance with MIL-STD-130.

3.6.2 Handling and electrical. Markings pertaining to the following characteristics and which conform to MIL-STD-1472, shall be applied to trainer:

Center of gravity	Location of:
Gross weight	Lift eyes
Electrical:	Fork lift bearings
Designations	Jacking points
Ratings	Casters
Safety precautions:	Towing attachments
Electrical	
Handling	

3.6.3 Decals. Decals conforming to Drawings 12265600 and 12277801 shall be applied to trainer as specified.

3.7 Workmanship. Workmanship shall be of such quality as to assure that trainers furnished under this specification are free from defects that compromise, limit, or reduce performance in

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intended use (see 6.1). Trainer shall be free from burrs, scratches, sharp edges, chips, corrosion, scale, dirt, spattered weld or solder, cracked insulation, loose connections, and loose fasteners.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Contractor's record examination. To determine conformance to 3.2, 3.3.2, 3.3.14, and 3.3.16, contractor's records shall be examined.

4.2 Classification of inspections. Inspections shall be classified as follows:

a. First article (initial production) inspection (see 4.4).
b. Quality conformance inspection (see 4.5).

4.3 Inspection conditions.

4.3.1 Machine gun location. Machine gun shall remain stowed during operation of other subassemblies.

4.3.2 Inspection site environment. Unless otherwise specified, inspection shall be conducted in environment prevailing at time and place of inspection.

4.3.3 Stanchions and safety chain. Stanchions and safety chain shall be removed during operation of turret.

4.4 First article (initial production) inspection. On beginning production, two units shall pass first article (initial production) inspection. One unit shall pass first production unit inspection (see 4.4.1), and one unit shall pass initial production inspection (see 4.4.2).

4.4.1 First production unit inspection.

4.4.1.1 In-process examination. During fabrication of the first production unit, an in-process examination will be conducted by representatives of TACOM Product Assurance Directorate to evaluate conformance of materials and workmanship to specified requirements. Examination shall be made at the contractor's, or subcontractor's, facility prior to application of primer and paint. Processing and welding procedures, quality system, and inspection records will be evaluated during this examination.

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4.4.1.2 Completed first production unit: contractor inspection. The first completed production unit shall be inspected by the contractor to determine conformance to contract and specifications. After inspection, the contractor shall submit acceptable unit (and all inspection records and certifications) to the responsible Government inspection element at contractor's plant for preliminary examination.

4.4.1.3 Preliminary examination. The responsible Government inspection element will conduct a preliminary examination, as specified in 4.5.2, of first completed production unit.

4.4.1.4 Provisional inspection. Provisional inspection of first completed production unit will be conducted jointly by representatives of TACOM Product Assurance Directorate and the responsible Government inspection element. Contractor shall provide any required assistance. Testing shall be as specified in 4.5.3. Contractor shall make available his inspection plan, records, and certifications pertinent to the unit and subassemblies.

4.4.1.5 Repair of defects. Defects found as a result of the foregoing inspections shall be corrected by the contractor at no cost to the Government. Failure of the contractor to correct defects promptly shall be cause for suspension of acceptance of units until corrective action has been accomplished and approved by the Government.

4.4.1.6 Unit disposition. On completion of first production unit inspection, the unit shall remain at the manufacturing facility as a production sample, and shall be the last unit shipped on contract. Unit may be released sooner at the discretion of the Government. The contractor shall service and maintain unit during this period.

4.4.1.7 Final approval and acceptance. Final approval and acceptance of first production unit shall be withheld until the second unit is accepted (see 4.4.2.1).

4.4.2 Second initial production unit inspection. One additional unit (see 4.4) from the first month's production or the first 20 produced shall be subjected to initial production inspection.

4.4.2.1 Initial production inspection. To determine conformance to Section 3 (inclusive), trainer shall be examined as specified in 4.5.2, and tested as specified in table IV. Inspection will be performed by the Government and will require no more than 90 days. The contractor shall expeditiously furnish repair parts as required to support testing at no cost to the Government.

4.4.2.2 Failure. Failure of the trainer to meet any requirement of this specification shall be cause for rejection of the trainer.

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TABLE IV. Classification of tests.

Characteristics	Requirement	Test	First article	Acceptance
Strength and rigidity	3.4.1	4.6.1	X	
Lift eye hoist	3.4.2	4.6.2	X	
Fork lift transport	3.4.3	4.6.3	X	
Towing attachments	3.4.4	4.6.4	X	
Casters	3.4.5	4.6.5	X	
Stow supports and brackets	3.4.6	4.6.6	X	
Seats	3.4.7	4.6.7	X	
Screw jacks and jack stands	3.4.8	4.6.8	X	
Access ladder	3.4.9	4.6.9	X	X
Stanchions and safety chains	3.4.10	4.6.10	X	X
Electrical operation	3.4.11	4.6.11	X	X
Current surge	3.4.12	4.6.12	X	
Steady state voltage	3.4.13	4.6.13	X	
Electromagnetic interference characteristics	3.4.14	4.6.14	X	X
Safety interrupts and override controls	3.4.15	4.6.15	X	X
Launcher fire/no-fire zones	3.4.15.1	4.6.15.1	X	X
Firing position indication	3.4.15.2	4.6.15.2	X	X
Loading position indication	3.4.15.3	4.6.15.3	X	X
Stow position indication	3.4.15.4	4.6.15.4	X	X
Intercom stations	3.4.16	4.6.16	X	X
Hydraulic pump noise	3.4.16.1	4.6.16.1	X	X
M60 machine gun mount	3.4.17	4.6.17	X	X
Locking device	3.4.17.1	4.6.17.1	X	X
Azimuth drive speeds				
Slew rate	3.4.18.1	4.6.18.1	X	X
Tracking rate	3.4.18.2	4.6.18.2	X	X
Elevation drive speed - Launcher	3.4.19	4.6.19	X	X
Minimum speed	3.4.19.1	4.6.19.1	X	X
Erection drive speed - Launcher	3.4.20	4.6.20	X	X

4.4.2.3 Unclassified defects. Defects considered to be departures from good workmanship but having no bearing on function, safety, interchangeability, or life shall be noted in writing. Such defects recurring in five consecutive lots, or ten lots or more within a 30 day period, will be added to the minor classification of defects with no increase in AQL (see 3.7, 6.4.1 and 6.4.2).

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4.5 Quality conformance inspection.

4.5.1 Sampling. Each turret trainer produced (100 percent) shall be subjected to examination (see 4.5.2) and acceptance tests (see 4.5.3).

4.5.2 Quality conformance examination.

4.5.2.1 Acceptable quality level. Each turret trainer shall be examined for conformance to following acceptable quality levels (AQL's) on the basis of percent defective:

Classification	AQL
Major	0.25
Minor	1.50

4.5.2.2 Classification of defects. For examination purposes, defects shall be classified according to MIL-STD-105, and as follows:

TABLE V. Classification of defects.

Categories	Defects	Method
Major:		
101	Dimensions affecting interchangeability not within tolerance (see 3.3. and 3.3.3).	Visual/SIE 1/
102	Mountings missing, improper, broken, bent (see 3.3.3, 3.3.3.1).	Visual/SIE
103	Level sensors missing, improperly installed, broken (see 3.3.3.2).	Visual/SIE
104	Electrical junction box incomplete, improper electrical properties, improper ground (see 3.3.3.3).	Visual/SIE
105	Power cable missing, incomplete, exposed conductor, improper insulation (see 3.3.3.4).	Visual
106	Turret improperly installed, inoperable (see 3.3.4).	Visual/SIE
107	Missile launcher improperly installed, inoperable (see 3.3.5).	Visual/SIE
108	Missile launcher mounting kit improperly installed (see 3.3.6).	Visual/SIE
109	Squad leader's periscope improperly installed, inoperable (see 3.3.7).	Visual/SIE
110	M26 periscope improperly installed, inoperable (see 3.3.8).	Visual/SIE

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TABLE V. Classification of defects. - Continued.

Categories	Defects	Method
111	M60 machine gun missing, improperly installed, inoperable (see 3.3.9).	Visual/SIE
112	Grenade launcher arm/fire box improperly installed, inoperable (see 3.3.10).	Visual/SIE
113	Cargo door improperly installed (see 3.3.11).	Visual/SIE
114	Intercom equipment missing, improperly installed, inoperable (see 3.3.12).	Visual/SIE
115	Welding missing, improper, broken, bent, misaligned (see 3.3.13).	Visual/SIE
116	Threaded parts missing, incorrect form, number per inch, class of fit, improper torque (see 3.3.15).	Visual/SIE
Minor:		
201	Dimensions not affecting interchangeability not within tolerance (see 3.3. and 3.3.3).	Visual/SIE
202	Walls improper, improper fit (see 3.3.3.1).	Visual
203	Finish improper, improperly applied (see 3.5.1, 3.5.2).	Visual/SIE
204	Marking missing, improper, improperly applied (see 3.6.1, 3.6.2, 3.6.3).	Visual
205	Workmanship inadequate, improper (see 3.7).	Visual

1/ = Standard inspection equipment.

4.5.2.3 Unclassified defects. Defects considered to be departures from good workmanship but having no bearing on function, safety, interchangeability, or life shall be noted in writing. Such defects recurring in five consecutive lots, or ten lots or more within a 30 day period, will be added to the minor classification of defects with no increase in AQL (see 3.7, 6.4.1 and 6.4.2).

4.5.2.4 Examination failure. If a trainer fails to pass any examination specified herein, the Government will stop acceptance until evidence has been provided by the contractor that corrective action has been taken. Failure of Government supplied equipment shall not be the responsibility of the trainer manufacturer unless failure is due to faulty installation.

4.5.3 Acceptance tests. Each trainer (100 percent, see 4.5.1), shall be examined as specified in 4.5.2 and tested as specified in table IV.

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4.5.3.1 Failure. Failure of trainer to conform to requirements of this specification shall be cause for rejection of failed unit.

4.5.3.2 Repair and retest. Trainers that have been rejected may be repaired or reworked and subjected again to examination and testing as specified in 4.5.2 and 4.5.3.

4.6 Methods of inspection.

4.6.1 Strength and rigidity test. To determine conformance to 3.4.1, trainer shall be completely assembled. Trainer shall be canted (tilted) so that vertical centerline forms an angle of 10° with perpendicular as observed on properly installed inclinometer. With appropriate crew aboard, each subassembly shall be operated throughout its full range.

4.6.2 Lift eye hoist test. To determine conformance to 3.4.2, trainer shall be connected to hoist by means of cables attached to all lift eyes and shall be raised from the ground, transported to another place, and lowered to the ground. Trainer shall then be measured to detect distortion and examined to detect broken or deformed parts.

4.6.3 Fork lift transport test. To determine conformance to 3.4.3, trainer shall be raised, transported to another place, and lowered to the ground by fork lift truck with forks applied to proper bearing points. Trainer shall then be measured to detect distortion and examined to detect broken or deformed parts.

4.6.4 Towing test. To determine conformance to 3.4.4, trainer with casters properly installed, shall be towed from either end over smooth, hard surface by prime mover connected to towing attachment. Trainer shall be measured to detect distortion and examined to detect broken or deformed parts. Towing for a substantial distance is not intended. Trainer is intended to be movable on its casters for convenience only.

4.6.5 Caster test. To determine conformance to 3.4.5, caster operation during procedure specified in 4.6.4, shall be observed to detect failure to roll or turn.

4.6.6 Stow supports and brackets test. To determine conformance to 3.4.6, launcher shall be aligned with and rest on stow supports and brackets. During and after test, supports and brackets shall be examined to detect broken or deformed parts.

4.6.7 Seat performance test. To determine conformance to 3.4.7, weights simulating crew shall be placed on seats as in intended use and service.

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4.6.8 Screw and jacks stand performance test. To determine conformance to 3.4.8, screw jacks and jack stands shall be installed on trainer and shall be adjusted until trainer is supported laterally at 10° angle from perpendicular. Subsequently, adjustments shall be made until trainer is supported longitudinally at 10° angle from perpendicular. Trainer, jacks, and jack stands shall be examined to detect broken or deformed parts.

4.6.9 Access ladder test. To determine conformance to 3.4.9, access ladder shall be placed in its vertical supports and scaled. Thereafter, ladder shall be removed from vertical anchorage and properly stowed. Ladder, supports, stow supports, and securing devices shall be examined to detect broken or deformed parts.

4.6.10 Stanchion and safety chain test. To determine conformance to 3.4.10, weights shall be applied to stanchions and safety chains to simulate falling personnel. Stanchions and chains shall be examined to detect broken or deformed parts.

4.6.11 Electrical operation test. To determine conformance to 3.4.11, trainer shall be electrically connected to 28 Vdc, 200 A power supply and tested as specified in 4.6.12 through 4.6.16.1.

4.6.12 Current surge test. To determine conformance to 3.4.12, junction box shall be electrically connected to proper power supply and to apparatus to apply and measure power and time. After application of 700 A for one second, electrical system shall be subjected to tests specified in 4.6.13 through 4.6.16.1.

4.6.13 Steady state voltage test. To determine conformance to 3.4.13, junction box shall be electrically connected to proper power supply and to apparatus to apply, display, and measure power and frequency. Power shall be applied and ripple voltage shall be observed as frequency is varied.

4.6.14 Electromagnetic interference characteristics test. To determine conformance to 3.4.14, trainer shall be subjected to applicable test procedures specified in MIL-STD-462.

4.6.15 Safety interrupts and override controls test. To determine conformance to 3.4.15, junction box shall be electrically connected to proper power supply. Master power switch shall be closed to pass power to all circuits. Controlled functions (see table II) shall be observed in connection with tests specified in 4.6.15.1 through 4.6.16.1. Variations from specifications shall be recorded.

4.6.15.1 Launcher fire/no-fire zones. To determine conformance to 3.4.15.1, launcher shall be raised to ERECT position and rotated while varying elevation angle. Indicator lamps and alarm shall be observed for proper operation specified in tables II and III.

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4.6.15.2 Firing position indicator test. To determine conformance to 3.4.15.2, erection arm and launcher assembly shall be moved to firing position and all hatches closed. Control panel indicator lamps shall be observed.

4.6.15.3 Loading position indicator test. To determine conformance to 3.4.15.3, erection arm and launcher assembly shall be moved to loading position. Control panel indicator lamps shall be observed.

4.6.15.4 Stow position indicator test. To determine conformance to 3.4.15.4, erection arm and launcher shall be moved to stow position. Control panel indicator lamps shall be observed.

4.6.16 Intercom stations test. To determine conformance to 3.4.16, intercom shall undergo following procedure:

a. Operate azimuth drive throughout 360° azimuth.
b. Operate any pair of intercom stations.
c. Observe and record any discontinuity in audio.

4.6.16.1 Hydraulic pump noise test. To determine conformance to 3.4.16.1, intercom shall undergo following procedure:

a. Operate azimuth drive.
b. Operate hydraulic pump.
c. Operate any pair of intercom stations.
d. Observe and record noise levels in audio attributable to hydraulic pump.

4.6.17 M60 machine gun mount test. To determine conformance to 3.4.17, machine gun mount shall be moved in connection with apparatus permitting observation of effort required. Effort shall be recorded and compared to requirement.

4.6.17.1 Locking device test. To determine conformance to 3.4.17.1, locking device shall be set and mount shall be moved in connection with apparatus permitting observation of effort required. Effort shall be recorded and compared to requirement.

4.6.18 Azimuth drive speeds.

4.6.18.1 Slew rate test. To determine conformance to 3.4.18.1, turret shall be rotated in connection with apparatus to permit observation of speed in terms of degrees per second. Speed shall be recorded and compared to requirement.

4.6.18.2 Tracking rate test. To determine conformance to 3.4.18.2, turret shall be rotated in connection with apparatus to permit observation of speed in terms of mrad per second. Speed shall be recorded and compared to requirement.

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4.6.19 Elevation drive speed test. To determine conformance to 3.4.19, launcher elevation drive shall be operated in conjunction with apparatus suitable for observing speed in terms of degrees per second. Speed shall be recorded and compared to requirement.

4.6.19.1 Elevation minimum speed test. To determine conformance to 3.4.19.1, elevation drive shall be operated in conjunction with apparatus suitable for observing speed in terms of mrad per second. Speed shall be recorded and compared to requirement.

4.6.20 Erection drive speed test. To determine conformance to 3.4.20, launcher erection drive shall be operated in conjunction with apparatus suitable for observing time of operation from stowed position to erect-locked position. Time required shall be observed and recorded for comparison with requirement.

5. PACKAGING

5.1 Unit processing. Turret trainers shall be processed for shipment and storage to extent indicated on unit preservation data sheets or other implementing document furnished by the responsible Government activity at the time of invitation for bids (see 6.2).

6. NOTES

6.1 Intended use. Turret trainer is intended for use in training military personnel in the preparation, use, and maintenance of turret, armament, and fire control devices installed on military combat vehicles.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Government furnished equipment (see 3.3.1).
c. Applicable drawing, specification, or standard for welding (see 3.3.13).
d. Unit preservation data sheets or other packaging documents (see 5.1).

6.3 First article. First article samples shall be tested and approved under the appropriate provisions of 7-104.55 of the Defense Acquisition Regulation. The contracting officer should include specific instructions in all acquisition instruments regarding arrangements for examination, tests, and approval of the first article (see 3.1).

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6.4 Deficiency sheet. The Government inspector shall verify that a thorough inspection of each turret trainer is performed by the contractor for the listed characteristics and departures from good workmanship. The Government inspector shall assure that defects encountered during inspection are enumerated on deficiency sheet for the turret trainer. Defects noted on a Deficiency Sheet shall contain sufficient description to enable the Government inspector and the contractor's representative to classify the defects in accordance with the classification of defects (see table V) and definitions contained in MIL-STD-105. Corrective action shall be taken for recurring defects.

6.4.1 Recurring major defects. A major defect is recurring when the same defect occurs more than once in the same sample or when the defect occurs in two successive samples. A major defect may be considered recurring when the historical inspection records ("P" chart or equivalent) reflect such a condition. Recurring major defects shall be cause for the entire lot or lots to be inspected for the recurring defects.

6.4.2 Recurring minor defects. A minor defect is recurring if it occurs more than twice in the same sample or when the defect occurs in four successive samples. Recurring minor defects shall be cause for the entire lot or lots to be inspected for the recurring defects.

6.5 Recycled materials. The use of recycled materials which meet the requirements of the applicable material specifications without jeopardizing the intended use of the item shall be encouraged (see 3.2).

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